Sample Paper 06

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

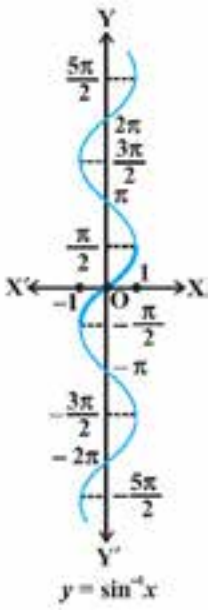
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the domain of the function $y = \sin^{-1}x$ as represented in the graph?



(a) $(-\infty,\infty)$ (b) $[0,\pi]$

(c) $[-1,1]$ (d) $[-\pi,\pi]$

2. Two dice are thrown n times in succession. The probability of obtaining a doublet six atleast once is

(a) $\left(\frac{1}{36}\right)^n$ (b) $1-\left(\frac{35}{36}\right)^n$

(c) $\left(\frac{1}{12}\right)^n$ (d) None of these

3. The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2+\left(\frac{dy}{dx}\right)^2 = x\sin\left(\frac{d^2y}{dx^2}\right)$

(a) 1 (b) 3

(c) 2 (d) none of these

4. The solution of the differential equation $x^2\frac{dy}{dx} - xy = 1+\cos\frac{y}{x}$ is

(a) $\tan\frac{y}{2x} = c-\frac{1}{2x^2}$ (b) $\tan\frac{y}{x} = c+\frac{1}{x}$

(c) $\cos\frac{y}{x} = 1+\frac{c}{x}$ (d) $x^2 = (c+x^2)\tan\frac{y}{x}$

5. Let G be the centroid of a triangle ABC and O be any other point, then $\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}$ is equal to

(a) $\vec{0}$ (b) $\overrightarrow{OG}$

(c) $3\overrightarrow{OG}$ (d) none of these

6. The projections of a line segment on x, y, z axes are 12, 4, 3. The length and the direction cosines of the line segment are

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(a) $13, < \frac{12}{13}, \frac{4}{13}, \frac{3}{13} >$ (b) $19, < \frac{12}{19}, \frac{4}{19}, \frac{3}{19} >$

(c) $11, < \frac{12}{11}, \frac{14}{11}, \frac{3}{11} >$ (d) none of these

7. If A and B are mutually exclusive events with $P(B) \neq 1$, then $P(A/\overline{B})$ is equal to (here, $\overline{B}$ is the complement of the event B)

(a) $\frac{1}{P(B)}$ (b) $\frac{1}{1-P(B)}$

(c) $\frac{P(A)}{P(B)}$ (d) $\frac{P(A)}{1-P(B)}$

8. Equation to the curve through (2,1) whose slope at the point (x, y) is $\frac{x^2+y^2}{2xy}$, is

(a) $2(x^2-y^2)=3x$ (b) $2(y^2-x^2)=6y$

(c) $x(x^2-y^2)=6$ (d) none of these

9. If $x = e^{y+e^{y+e^{y+\cdots}}}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{1}{x}$ (b) $\frac{1-x}{x}$

(c) $\frac{x}{1+x}$ (d) None of these

10. The point of discontinuous of $\tan x$ are

(a) $n\pi,\ n \in I$ (b) $2n\pi,\ n \in I$

(c) $(2n+1)\frac{\pi}{2},\ n \in I$ (d) None of these

11. The area bounded by the parabola $y^2 = 8x$ and its latus rectum is

(a) 16/3 sq units (b) 32/3 sq units

(c) 8/3 sq units (d) 64/3 sq units

12. If $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$, then $\vec{a}, \vec{b}$ are

(a) Perpendicular (b) Like parallel

(c) Unlike parallel (d) Coincident

13. Let $f(x) = x^3 + \frac{3}{2}x^2 + 3x + 3$, then $f(x)$ is

(a) Am even function

(b) An odd function

(c) An increasing function

(d) A decreasing function

14. If $P = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 1 \end{bmatrix}$ and $Q = PP^T$, then the value of Q is

(a) 2 (b) -2

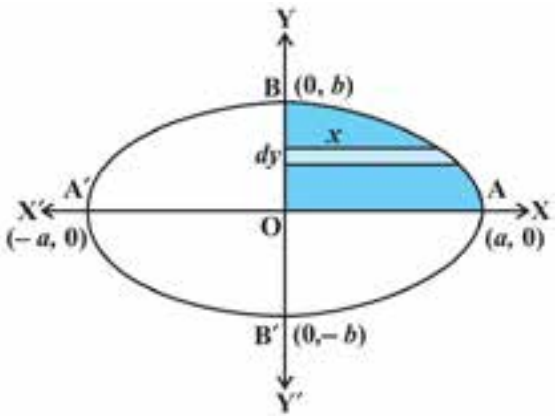
(c) 1 (d) 0

15. $\int \frac{dx}{x(x^7+1)}$ is equal to

(a) $\log\left(\frac{x^7}{x^7+1}\right)+C$ (b) $\frac{1}{7}\log\left(\frac{x^7}{x^7+1}\right)+C$

(c) $\log\left(\frac{x^7+1}{x^7}\right)+C$ (d) $\frac{1}{7}\log\left(\frac{x^7+1}{x^7}\right)+C$

16. In Fig., the shaded region lies inside which type of curve?



(a) Circle with radius a

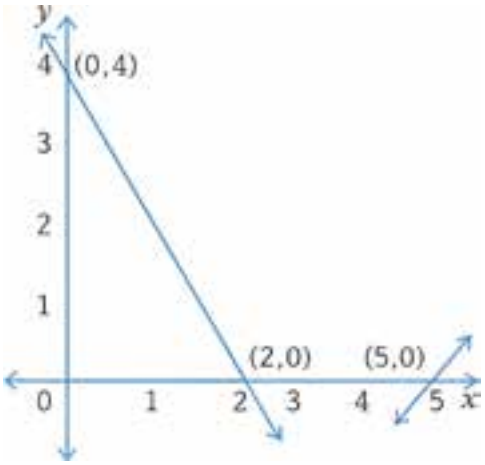
(b) Ellipse centered at O with semi-major axis a and semi-minor axis b

(c) Hyperbola with transverse axis on the X-axis

(d) Parabola opening along the Y-axis

17. Why does the minimum value of $Z = 3x + 2y$ exist for this problem even though the feasible region is unbounded?

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(a) Because the feasible region is bounded by all constraints.

(b) Because the inequality 3x+2y<6 has no point in common with the feasible region.

(c) Because the inequality 3x+2y<6 has infinitely many points in common with the feasible region.

(d) Because every LPP always has both a minimum and maximum solution.

18. The length of the largest interval in which the function $3\sin x - 4\sin^3 x$ is increasing, is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$ (d) π

19. **Assertion:** If $A = \begin{bmatrix} \sqrt{3} & 1 \\ 2 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} x & y & z \\ a & b & c \end{bmatrix}$, then orders of $(A + B)$ is 2×3

Reason: If $[aij]$ and $[bij]$ are two matrics of the same order, then order of $A + B$ is the same as the order of A or B

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. **Assertion:** The relation R in a set $A = \{1,2,3,4\}$ defined by $R = \{(x, y): 3x - y = 0)\}$ have the domain $= \{1,2,3,4\}$ and range $= \{3,6,9,12\}$

Reason: Domain & range of the relation (R) is respectively the set of all first & second entries of the distinct ordered pair of the relation.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. If a line makes angles 90° and 60°, respectively with the positive directions of X and Y-axes, find the angle which it makes with the positive direction of Z-axis.

22. Evaluate $\int \frac{2}{1+\cos 2x}\,dx$.

OR

Write the value of $\int \frac{x+\cos 6x}{3x^2+\sin 6x}\,dx$.

23. Prove that if E and F are independent events, then the events E and F' are also independent.

24. Find λ and μ, if $(\hat{i} + 3\hat{j} + 9\hat{k}) \times (3\hat{i} - \lambda\hat{j} + \mu\hat{k}) = \vec{0}$.

OR

If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then write the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.

25. If $R = \{(x, y): x + 2y = 8\}$ is a relation on N, then write the range of R.

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. If $y = e^{\tan^{-1}x}$, prove that

$$(1 + x^2)\frac{d^2y}{dx^2} + (2x - 1)\frac{dy}{dx} = 0.$$

27. If $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}$, write the value of $|AB|$.

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28. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a}+\vec{b}$ is also a unit vector, then find the angle between $\vec{a}$ and $\vec{b}$.

29. Write the value of $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.

30. Evaluate $\int \frac{\cos 2x-\cos 2\alpha}{\cos x-\cos\alpha}\,dx$.

OR

Evaluate $\int \frac{dx}{x(x^5+3)}$.

31. Find the intervals in which the function $f(x)=3x^4-4x^3-12x^2+5$ is

(i) strictly increasing

(ii) strictly decreasing.

OR

Find the intervals in which the function given by $f(x)=\frac{3}{10}x^4-\frac{4}{5}x^3-3x^2+\frac{36x}{5}+11$ is

(i) strictly increasing

(ii) strictly decreasing.

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. A wire of length 28 m is to be cut into two pieces. One of the two pieces is to be made into a square and the other into a circle. What should be the lengths of two pieces, so that the combined area of circle and square is minimum?

OR

Show that the volume of the largest cone that can be inscribed in a sphere of radius R is $\frac{8}{27}$ of the volume of the sphere.

33. Find $\int \frac{4}{(x-2)(x^2+4)}\,dx$.

OR

Find $\int \frac{2x}{(x^2+1)(x^2+2)^2}\,dx$.

34. An urn contains 4 white and 6 red balls. Four balls are drawn at random (without replacement) from the urn. Find the probability distribution of the number of white balls?

OR

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors of the same magnitude, then prove that $\vec{a}+\vec{b}+\vec{c}$ is equally inclined with the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

35. Find the angle between following pair of lines.

$\frac{-x+2}{-2}=\frac{y-1}{7}=\frac{z+3}{-3}$ and $\frac{x+2}{-1}=\frac{2y-8}{4}=\frac{z-5}{4}$

and check whether the lines are parallel or perpendicular.

OR

Find the shortest distance between lines whose vector equations are

$$\vec{r}=(1-t)\,\hat{i}+(t-2)\,\hat{j}+(3-2t)\,\hat{k}$$

$$\vec{r}=(s+1)\,\hat{i}+(2s-1)\,\hat{j}-(2s+1)\,\hat{k}.$$

SECTION - E

Case study based questions are compulsory.

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36. Global Air Lines has contracted with a tour group to transport a minimum of 1,600 first-class passengers and 4,800 economy-class passengers from New York to London during a 6-month time period. Global Air has two types of airplanes, the Orville 606 and the Wilbur W-1112. The Orville 606 carries 20 first-class passengers and 80 economy-class passengers and costs \$12,000 to operate. The Wilbur W-1112 carries 80 first-class passengers and 120 economy-class passengers and costs \$18,000 to operate.



During the time period involved, Global Air can schedule no more than 52 flights on Orville 606s and no more than 30 flights on Wilbur W-1112s.

(i) How should Global Air Lines schedule its flights to minimize its costs?

(ii) What operating costs would this schedule entail?

37. Chemical reaction, a process in which one or more substances, the reactants, are converted to one or more different substances, the products. Substances are either chemical elements or compounds. A chemical reaction rearranges the constituent atoms of the reactants to create different substances as products.



In a certain chemical reaction, a substance is converted into another substance at a rate proportional to the square of the amount of the first substance present at any time t. Initially $(t = 0)$ 50 g of the first substance was present; 1 hr later, only 10 g of it remained.

(i) Find an expression that gives the amount of the first substance present at any time t.

(ii) What is the amount present after 2 hr?

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38. Rice is a nutritional staple food which provides instant energy as its most important component is carbohydrate (starch). On the other hand, rice is poor in nitrogenous substances with average composition of these substances being only 8 per cent and fat content or lipids only negligible, i.e., 1 per cent and due to this reason it is considered as a complete food for eating. Rice flour is rich in starch and is used for making various food materials.



Two farmers Ramkishan and Gurcharan Singh cultivate only three varieties of rice namely Basmati, Permal and Naura. The sale (in ₹) of these varieties of rice by both the farmers in the month of September and October are given by the following matrices A and B.

September Sales (in ₹)

$$A = \begin{array}{c} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 10000 & 20000 & 30000 \\ 50000 & 30000 & 10000 \end{bmatrix} \end{array} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

October Sales (in ₹)

$$B = \begin{array}{c} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \end{array} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(i) Find the combined sales in September and October for each farmer in each variety.

(ii) Find the decrease in sales from September to October.

(iii) If both farmers receive 2% profit on gross sales, compute the profit for each farmer and for each variety sold in October.

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